COURSE MATERIAL	
SUBJECT	BASIC ELECTRICAL & ELECTRONICS ENGINEERING(EE23AES101)
UNIT	2
COURSE	B.TECH
DEPARTMENT	ECE
SEMESTER	1-1
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Version	V-1
PREPARED / REVISED DATE	08-03-2021

TABLE OF CONTENTS – UNIT 1

1. Course Objectives

The objectives of the course are to make the students learn about:

1. To introduce basics of electric circuits.
2. To teach DC and AC electrical circuit analysis.
3. To explain working principles of transformers and electrical machines.
4. To impart knowledge on low voltage electrical installations

2. Prerequisites

Students should have knowledge on

1. Mathematics.
2. Physics.

3. Syllabus

UNIT II DC & AC Machines

Construction and working Principle of DC Generator - EMF equations - OCC characteristics of DC generator – principle and operation of DC Motor – Performance Characteristics of DC Motor - Speed control of DC Motor – Construction and working Principle of Single Phase Transformer - OC and SC test on transformer - principle and operation of Induction Motor and Synchronous Generator

4. Course outcomes

1. **Recall** the properties of magnetic material (L2).
2. **Analyze** simple magnetic circuits with DC excitation (L4).
3. **Apply** network theorems to simple circuits (L3).
4. **Analyze** different types of dc and ac machines (L4).

5. Co-PO / PSO Mapping

Machine Tools	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1			3	3										
CO2			3	3										
CO3			3	3										
CO4			3	3										

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Introduction	T1, R1
2		INTRODUCTION	T1, R1
3		CONSTRUCTIONAL DETAILS OF DC MACHINES	T1, R1
4		WORKING PRINCIPLE AND OPERATION OF DC GENERATOR	T1, R1
5		EMF EQUATION OF A DC GENERATOR	T1, R2
6		PROBLEMS	T1, R1
7	2	WORKING PRINCIPLE AND OPERATION OF A DC MOTOR	T1, R1

8		METHODS OF SPEED CONTROL OF DC MOTOR	T1, R1
9		PERFORMANCE CHARACTERISTICS OF DC MOTORS	T1, R1
10		APPLICATIONS	T1, R1
11	3	THREE PHASE INDUCTION MOTOR	T1, R1
12		SINGLE PHASE TRANSFORMER	T1, R1
13		SYNCHRONOUS GENERATOR	T1, R1
14		APPLICATIONS	T1, R1

7. Activity Based Learning

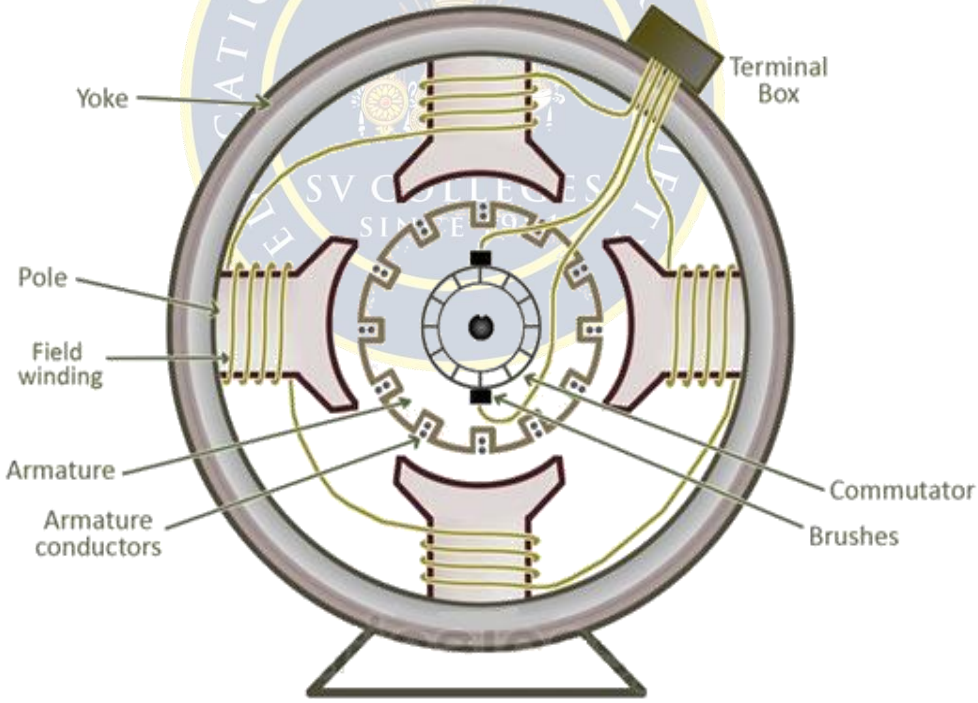
1. Explaining DC and AC machines in laboratory for practical knowledge.
2. Mini project on different machines..

8. Lecture Notes

2.1 INTRODUCTION

DC Machines are types of electrical machines that use dc current in the case of dc motors and generates dc voltages in case of dc generator. DC motor transforms electrical power into mechanical power and the generator converts mechanical power into electrical.

2.2 CONSTRUCTIONAL DETAILS OF DC MACHINE



2.2.1 YOKE

Yoke is also called as Frame. Main function of the Yoke is to protect the internal parts of machine from any mechanical injury, dust or moisture. It provides mechanical support to the machine. It also provides the passage for magnetic flux produced by the poles. For small machines, yoke is made up of Cast Iron and for large machines it is made up of fabricated steel.

2.2.2 POLE AND POLE SHOE

Pole of a DC machine is like an Electromagnet. Field winding is placed on it and produces the magnetic flux. Poles are made up of thin cast iron lamination riveted together. Purpose of the pole shoe is to enlarge the cross section area so that the reluctance of the magnetic path is reduced. It spreads the magnetic flux in the air gap more uniformly.

1.2.3 FIELD WINDING

Purpose of the field winding is to produce the magnetic flux when an electric current is passed through it. It is placed on pole and a small DC source is connected to it. Material used for Field Winding is Enameled Copper Wire.

1.2.4 ARMATURE

Armature is the rotating part of the machine and is cylindrical in shape. It is made up of thin silicon steel lamination, which is circular in shape and are riveted together. Thin lamination is used to reduce Eddy Current Loss. On the outer periphery/circumference of the armature, slots are provided to accommodate the Armature winding.

1.2.5 ARMATURE WINDING

They are placed in the slots provided on Armature. Made up of enameled copper wire and have multi-turns. When the Armature rotates, it will also rotate and an emf is induced in these winding.

1.2.5 COMMUTATOR

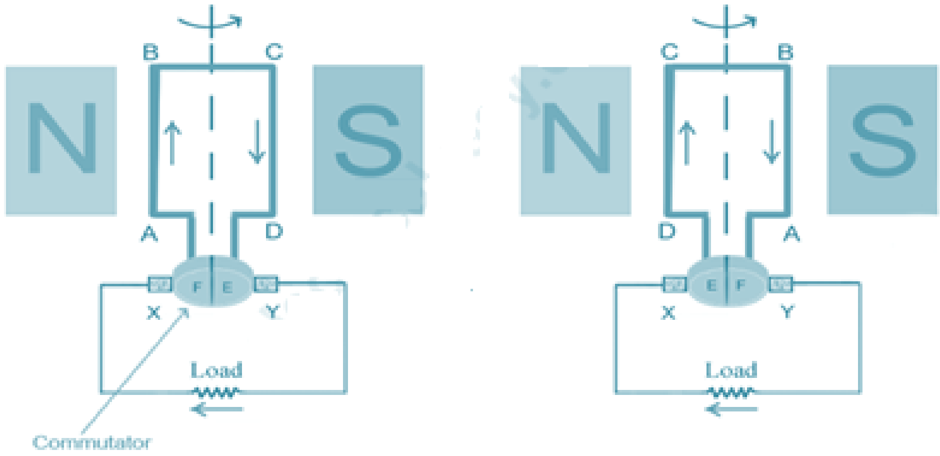
It is mounted on the same shaft as the Armature. Its function is to connect the rotating Armature winding to the stationary external circuit using brushes. It also provides uni-directional torque in case of DC motor. Commutator is cylindrical in shape and is made up of hard drawn copper segments. These segments are insulated from each other by a thin sheet of mica.

1.2.5 BRUSHES

Brushes are placed and pressed upon the commutator and makes a connecting link between the armature winding and the external circuit. Brushes are made up of high grade carbon and it is placed in a particular position around commutator by brush holder.

2.3 WORKING PRINCIPLE AND OPERATION OF A DC GENERATOR

According to Faraday's laws of electromagnetic induction, whenever a conductor is placed in a varying magnetic field, an emf gets induced in the conductor. The magnitude of induced emf can be calculated from the emf equation of dc generator. If the conductor is provided with a closed path, the induced current will circulate within the path. In a DC generator, field coils produce an electromagnetic field and the armature conductors are rotated into the field. Thus, an electromagnetically induced emf is generated in the armature conductors. The direction of induced current is given by Fleming's right hand rule.



Case 1 Case 2

The right hand is held with the thumb, index finger and middle finger mutually perpendicular to each other (at right angles), the thumb is pointed in the direction of the motion of the conductor relative to the magnetic field, the middle finger indicates the direction of magnetic field and index finger indicates the direction of induced current.

According to Fleming's right hand rule, the direction of induced current changes whenever the direction of motion of the conductor changes. Let's consider an armature rotating clockwise and a conductor at the left is moving upward. When the armature completes a half rotation, the direction of motion of that particular conductor will be reversed to downward. Hence, the direction of current in every armature conductor will be alternating. If you look at the above figure, you will know how the direction of the induced current is alternating in an armature conductor. But with a split ring commutator, connections of the armature conductors also get reversed when the current reversal occurs. And therefore, we get unidirectional current at the terminals.

2.4 EMF EQUATION OF DC GENERATOR

Let,

Φ = Flux per pole in Weber.

Z = Total number of armature conductors.

N = Armature rotation in revolution per minute (r.p.m).

P = Number of poles.

A = Number of parallel paths in armature.

E = e.m.f induced in any parallel path or generated e.m.f.

According to Faraday's law of Electromagnetic induction.

Average e.m.f generated per conductors,

$$= \frac{d\phi}{dt} = \frac{flux\ cut}{time\ taken}\ volt$$

Flux cut per Conductors in one revolution, dφ = φ P wb

Number of revolutions per minute = N
Time taken for one revolution, dt = 60 / N sec
E.m.f generated per conductor,

$$= \frac{d\phi}{dt} = \frac{\phi P}{60 / N} \times \frac{\phi PN}{60}\ volt$$

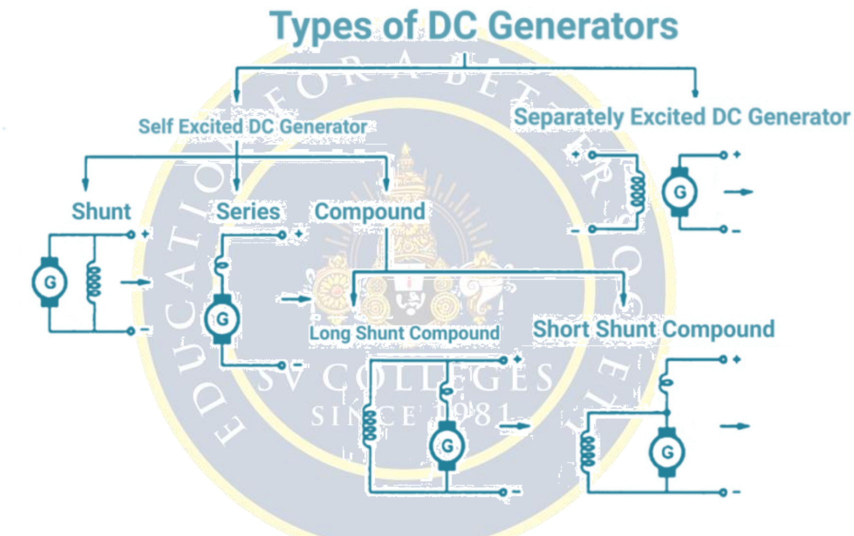
Therefore, the total emf E generated between the terminals if given as,
E = (Average e.m.f generated per conductor)*(Number of conductor in each parallel path)

$$E = \frac{\phi PN}{60} \times \frac{Z}{A}\ volt$$

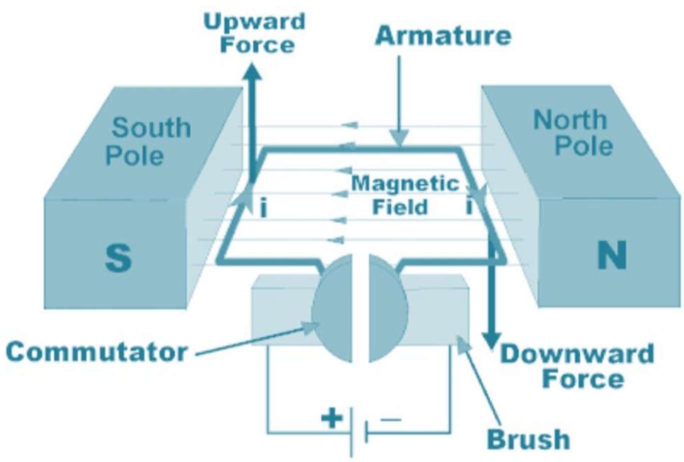
$$E\ = \frac{\phi PN}{60} \times \frac{Z}{A}\ volt$$

Where, A = P for lap winding. A = 2 for wave winding

2.4.1 TYPES OF DC GENERATORS



2.5 WORKING PRINCIPLE AND OPERATION OF A DC MOTOR



An electric motor is an electrical machine which converts electrical energy into mechanical energy. The basic working principle of a DC motor is: "whenever a current carrying conductor is placed in a magnetic field, it experiences a mechanical force". The direction of this force is given by Fleming's left-hand rule and its magnitude is given by F = BIL. Where, B = magnetic flux density, I = current and L = length of the conductor within the magnetic field.

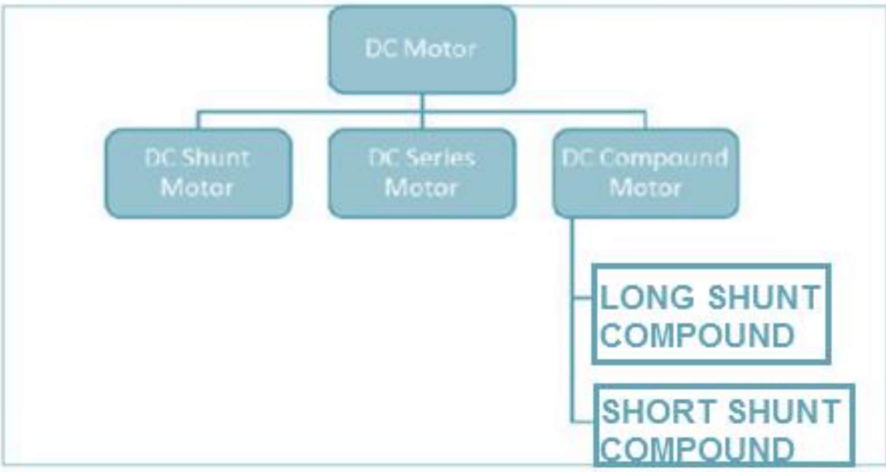
Fleming's left hand rule: If we stretch the first finger, second finger and thumb of our left hand to be perpendicular to each other, and the direction of magnetic field is represented by the index finger, direction of the current is represented by the middle finger, and then the thumb represents direction of the force experienced by the current carrying conductor.

When armature windings are connected to a DC supply, an electric current sets up in the winding. Magnetic field may be provided by field winding (electromagnetism) or by using permanent magnets. In this case, current carrying armature conductors experience a force due to the magnetic field, according to the principle stated above. Commutator is made segmented to achieve unidirectional torque. Otherwise, the direction of force would have reversed every time when the direction of movement of conductor is reversed in the magnetic field.

Back EMF According to fundamental laws of nature, no energy conversion is possible until there is something to oppose the conversion. In case of generators this opposition is provided by magnetic drag, but in case of dc motors there is back emf.

When the armature of a motor is rotating, the conductors are also cutting the magnetic flux lines and hence according to the Faraday's law of electromagnetic induction, an emf induces in the armature conductors. The direction of this induced emf is such that it opposes the armature current (Ia). The circuit diagram below illustrates the direction of the back emf and armature current. Magnitude of the Back emf can be given by emf equation of a DC generator.

2.5.1 TYPES OF DC MOTORS



2.6 METHODS OF SPEED CONTROL OF DC MOTOR

2.6.1. Armature Resistance Control of DC Motor
2.6.2. Flux Control Method of DC Motor
2.6.3. Voltage Control Method of DC Motor

2.6.1. Armature Resistance Control of DC Motor

SHUNT MOTOR

W.K.T $N = K\,Eb / \phi$

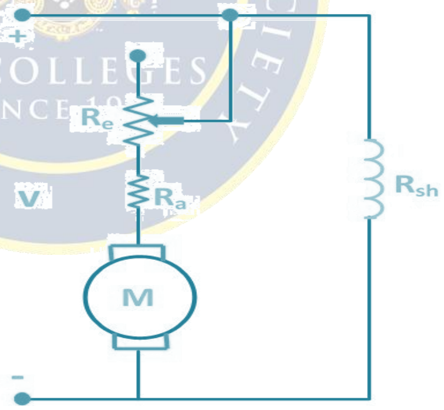
$N \propto Eb / \phi$

$N \propto (V - IaRa)/ \phi$, K is a constant

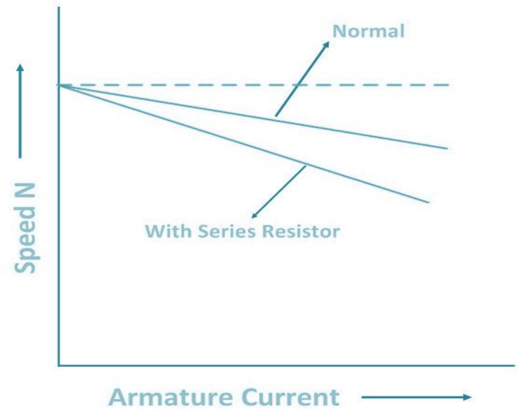
$N \propto (V - IaRa)$ ϕ is a constant for DC Shunt Generator

Speed of a dc motor is directly proportional to the back emf Eb and $Eb = V - IaRa$.

That means, when supply voltage V and the armature resistance Ra are kept constant, then the speed is directly proportional to armature current Ia. Thus, if we add resistance in series with the armature, Ia decreases and, hence, the speed also decreases. Greater the resistance in series with the armature, greater the decrease in speed. The connection diagram of a shunt motor of the armature resistance control method is shown below. In this method, a variable resistor Re is put in the armature circuit. The variation in the variable resistance does not affect the flux as the field is directly connected to the supply mains.



The speed current characteristic of the shunt motor is shown below.



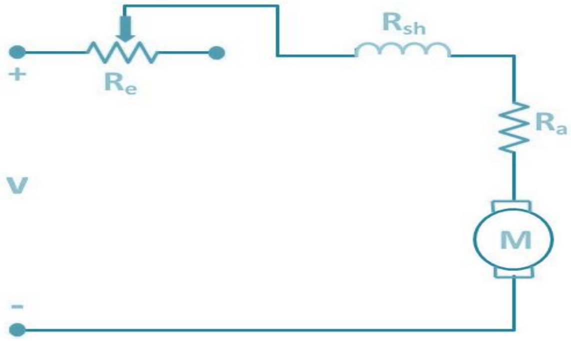
SERIES MOTOR

W.K.T $N = K\ Eb / \phi$

$N \propto Eb / \phi$

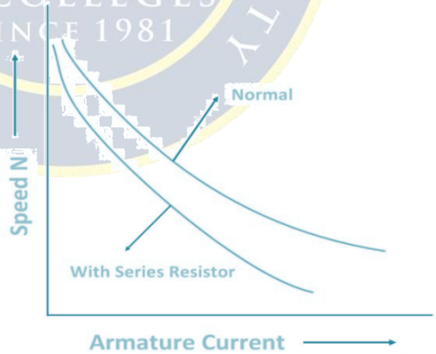
$N \propto (V - IaRa)/ \phi$, K is a constant

Now, let us consider a connection diagram of speed control of the DC Series motor by the armature resistance control method.



By varying the armature circuit resistance, the current and flux both are affected. The voltage drop in the variable resistance reduces the applied voltage to the armature, and as a result, the speed of the motor is reduced.

The speed–current characteristic of a series motor is shown in the figure below.



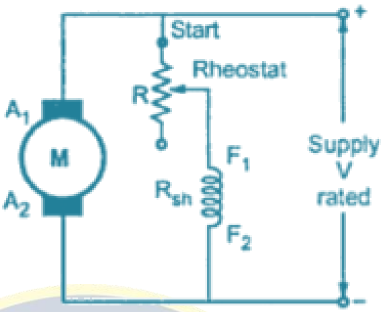
2.6.2. Flux Control Method of DC Motor

Flux is produced by the field current. Thus, the speed control by this method is achieved by control of the field current.

SHUNT MOTOR

We know that the speed of a dc motor is inversely proportional to the flux per pole. Thus by decreasing the flux, speed can be increased and vice versa.

To control the flux, a rheostat is added in series with the field winding, as shown in the circuit diagram. Adding more resistance (RC) in series with the field winding reduces the field current, and hence the flux is also reduced. This reduction in flux increases the speed, and thus, the motor runs at a speed higher than the normal speed. Therefore, this method is used to give motor speed above normal or to correct the fall of speed because of the load.



The speed–current characteristic for shunt motor is shown below:

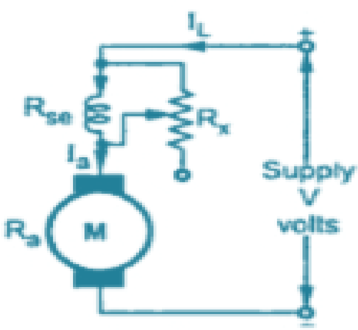
Fig. 2 N Vs I_{sh} (Φ) for shunt motor

SERIES MOTOR

In a series motor, the variation in field current is done by anyone method, i.e. either by a diverter or by a tapped field control.

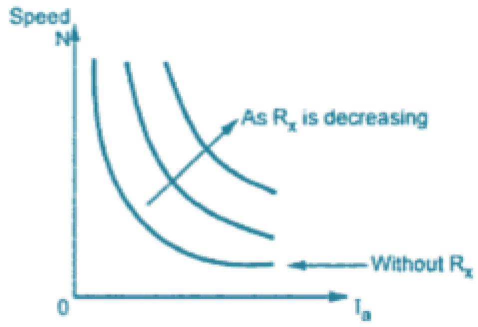
By Using a Diverter

A variable resistance Rx is connected in parallel with the series field windings as shown in the figure below:



(a) Field diverter

The parallel resistor is called a Diverter. A portion of the main current is diverted through a variable resistance Rx. Thus, the function of a diverter is to reduce the current flowing through the field winding. The reduction in field current reduces the amount of flux and as a result the speed of the motor increases. The speed–current characteristic for series motor is shown below.



(b) Speed-current characteristics

2.6.3 VOLTAGE CONTROL METHOD OF DC MOTOR

DC Shunt Motor - Multiple voltage control:

In this method, the shunt field is connected to a fixed exciting voltage and armature is supplied with different voltages. Voltage across armature is changed with the help of suitable switchgear. The speed is approximately proportional to the voltage across the armature.

DC series motor - Series-Parallel Control

This system is widely used in electric traction, where two or more mechanically coupled series motors are employed. For low speeds, the motors are connected in series, and for higher speeds, the motors are connected in parallel.

When in series, the motors have the same current passing through them, although voltage across each motor is divided. When in parallel, the voltage across each motor is same although the current gets divided.

2.7 PERFORMANCE CHARACTERISTICS OF DC MOTORS

Generally, three characteristic curves are considered important for DC motors which are,

(i) Torque vs. armature current,

(ii) Speed vs. armature current and

(iii) Speed vs. torque.

These characteristics are determined by keeping the following two relations in mind.

$Ta \propto \phi.Ia$ and $N \propto Eb/\phi$

For a DC motor, magnitude of the back emf is given by the same emf equation of a dc generator i.e. $Eb = P\phi NZ / 60A$. For a machine, P, Z and A are constant, therefore, $N \propto Eb/\phi$

2.7.1 Characteristics of DC Series Motors

1. Torque Vs. Armature Current (Ta-Ia)

We know that torque is directly proportional to the product of armature current and field flux, $Ta \propto \phi.Ia$. In DC series motors, field winding is connected in series with the armature, i.e. Ia = If. Therefore, before magnetic saturation of the field, flux ϕ is directly proportional to Ia. Hence, before magnetic saturation Ta a Ia2. Therefore, the Ta-Ia curve is parabola for smaller values of Ia.

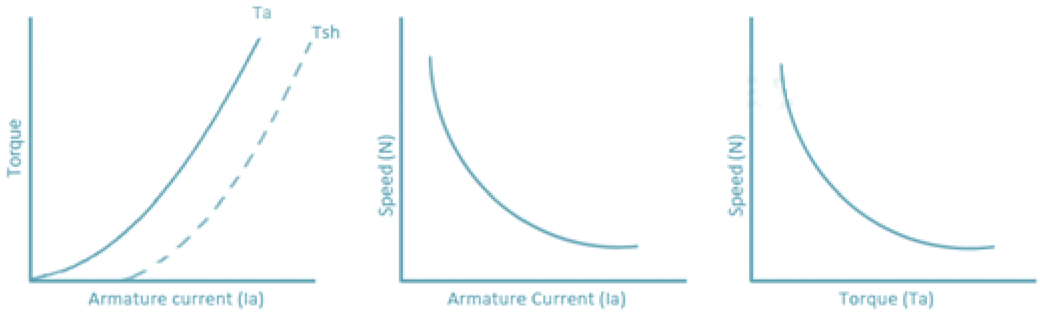
2. Speed Vs. Armature Current (N-Ia)

We know the relation, $N \propto Eb/\phi$

For small load current (and hence for small armature current) change in back emf Eb is small and it may be neglected. Hence, for small currents speed is inversely proportional to ϕ. As we know, flux is directly proportional to Ia, speed is inversely proportional to Ia. Therefore, when armature current is very small the speed becomes dangerously high.

3. Speed Vs. Torque (N-Ta)

From the above two characteristics of DC series motor, it can be found that when speed is high, torque is low and vice versa.



Characteristics of DC series motor

2.7.2 Characteristics of DC Shunt Motors

1. Torque Vs. Armature Current (Ta-Ia)

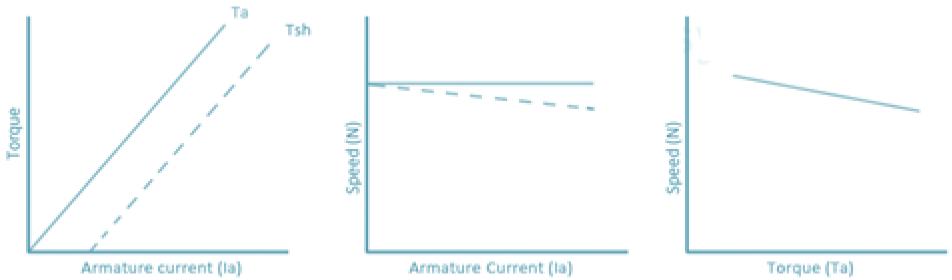
In case of DC shunt motors, we can assume the field flux ϕ to be constant. Though at heavy loads, ϕ decreases in a small amount due to increased armature reaction. As we are neglecting the change in the flux ϕ, we can say that torque is proportional to armature current. Hence, the Ta-Ia characteristic for a dc shunt motor will be a straight line through the origin.

2. Speed Vs. Armature Current (N-Ia)

As flux ϕ is assumed to be constant, we can say N ∝ Eb. But, as back emf is also almost constant, the speed should remain constant. But practically, ϕ as well as Eb decreases with increase in load. Back emf Eb decreases slightly more than ϕ, therefore, the speed decreases slightly.

3. Speed Vs. Torque (N-Ta)

From the above two characteristics of DC shunt motor, it can be found that the speed almost remains constant through torque changes from no load to full load conditions.



Characteristics of DC shunt motor

2.8 THREE PHASE INDUCTION MOTOR

The three-phase induction motor is a rotating electric machine that is designed to operate on a three-phase supply. This 3 phase induction motor is also called as an asynchronous motor. Three-phase induction motors are of two types: squirrel and slip-ring type induction motors.

PRINCIPLE OF OPERATION

When a balanced 3-phase excitation fed to stator the magnetic flux is setup with constant magnitude and rotating at synchronous speed.

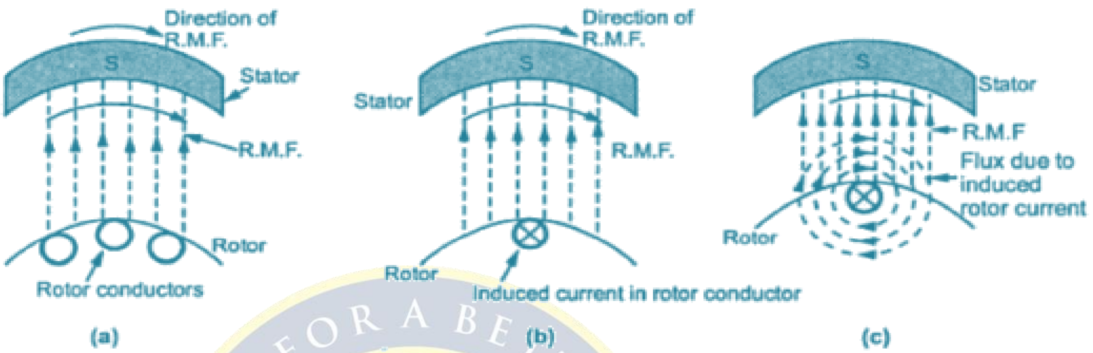
The flux passes through the air gap, cuts the rotor surface and rotor conductors which are at stationary.

Due to relative speed between the RMF and stationary rotor conductors, an emf is induced in the rotor according to faraday's law of electromagnetic induction.

At the starting, the frequency of induced emf is same as supply frequency.

Its magnitude is proportional to the relative speed between the flux and conductors, its direction given by FRHR.

The rotor bars (or) conductors framed a closed circuit, rotor current is produced whose direction is given by lens's law, is such very cause producing it.



Its cause is relative speed between RMF and stationary rotor conductors, effect is rotor current.

Hence in order to reduce the relative speed, the rotor starts running in the same direction as the flux and tries to catch up with rotation flux.

If rotor runs at synchronous speed in the direction of RMF, then there would be no flux cutting action. No emf in the rotor conductors. No current in the rotor parts and therefore no developed torque.

Hence the rotor of 3-phase induction motor can never attain synchronous speed.

Hence the relative speed difference between emf and rotor flux will not be zero and also the relative speed between stator and rotor field is called as slip. This slip is responsible for the induced emf in rotor and there by torque is produced.

2.9 SINGLE PHASE TRANSFORMER

Transformer is a static device that transfers electric power in one circuit to another circuit of the same frequency. It consists of primary and secondary windings. This transformer operates on the principle of mutual inductance.

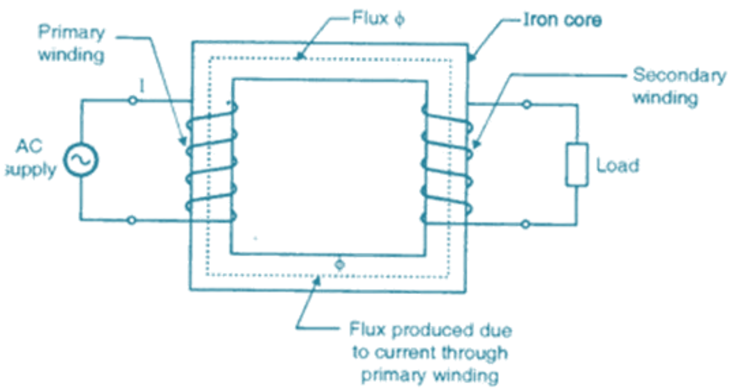


FIG: Single Phase Transformer Schematic

2.9.1 TRANSFORMER CONSTRUCTION

The three main parts of a transformer are:

Primary Winding: The winding that takes electrical power, and produces magnetic flux when it is connected to an electrical source.

Magnetic Core: This refers to the magnetic flux produced by the primary winding. The flux passes through a low reluctance path linked with secondary winding creating a closed magnetic circuit.

Secondary Winding: The winding that provides the desired output voltage due to mutual induction in the transformer.

2.9.2 WORKING PRINCIPLE OF TRANSFORMERS

The working principle of the single phase transformer is based on the Faraday's law of electromagnetic induction. Basically, mutual induction between two or more windings is responsible for transformation action in an electrical transformer. According to Faraday's law, "Rate of change of flux linkage with respect to time is directly proportional to the induced EMF in a conductor or coil".

The primary winding is supplied an alternating electrical source. The alternating current through the primary winding produces an alternating flux that surrounds the winding. Another winding, also known as the secondary winding, is brought close to the primary winding. Eventually, some portion of the flux in the primary will link with the secondary. As this flux is continually changing in amplitude and direction, there is a change in flux linkage in the second winding as well. According to Faraday's law of electromagnetic induction, an electromotive force (emf) is induced in the secondary winding which is called as induced emf. If the circuit of the secondary winding is closed an induced current will flow through it. This is the simplest form of electrical power transformation; this is the most basic working principle of a transformer.

The principle of operation of a transformer has been explained in the following simple steps:

- As soon as the primary winding is connected to a single–phase supply, an AC current starts flowing through it.
- An alternating flux is produced in the core by the AC primary current.
- The alternating flux gets linked with the secondary winding through the core.
- Now, according to Faraday's laws of electromagnetic induction this varying flux will induce voltage into the secondary winding.

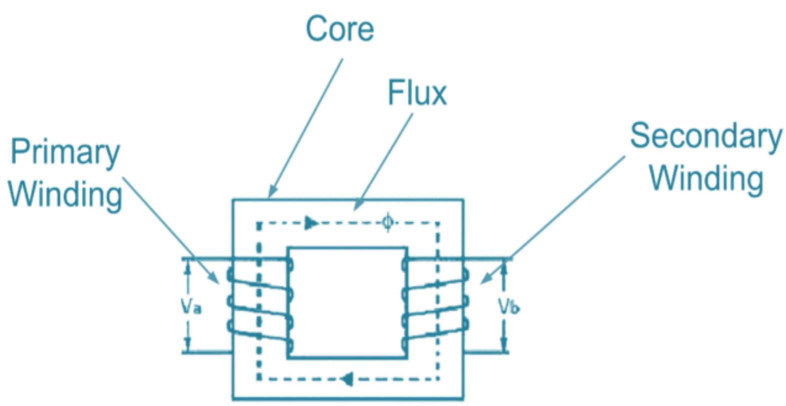
2.9.2 CONSTRUCTION DETAILS OF SINGLE PHASE TRANSFORMERS

Based on construction Transformers are two types

1. Core type
2. Shell type Transformer

1. CORE TYPE

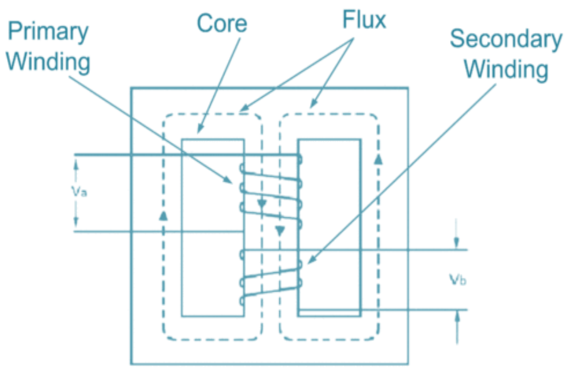
In core type transformers, winding is positioned on two limbs of the core and there is ONLY one flux path and windings are circumventing the core.



These transformers are quite favorite in High voltage practical applications like Distribution, Power, and Auto-Transformers.

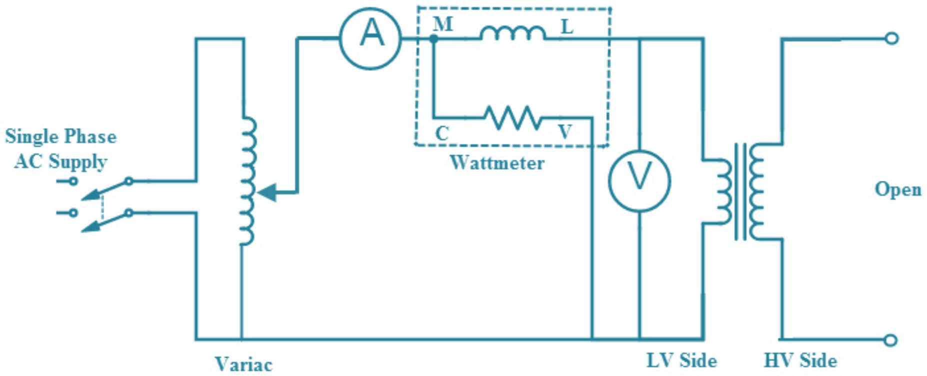
2. SHELL TYPE TRANSFORMER

In shell type transformers, winding is positioned on the middle limb of the core while other limbs are utilized as the mechanical support.



2.9.2 OPEN-CIRCUIT TEST ON SINGLE PHASE TRANSFORMER

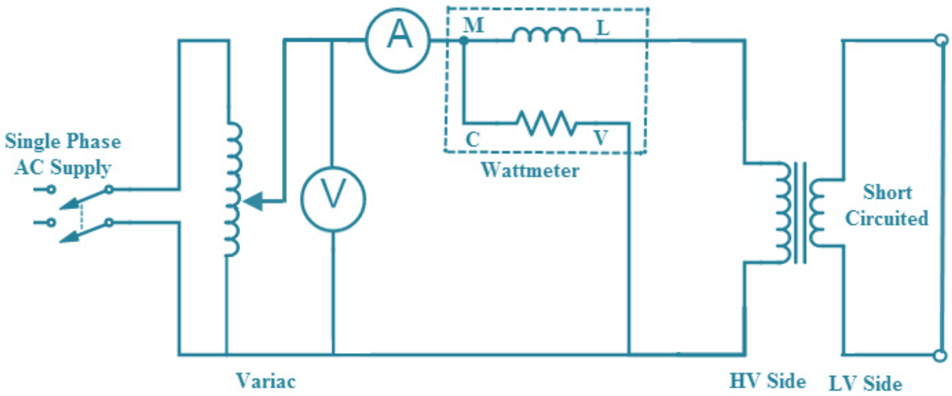
The connection diagram for open circuit test as shown in figure.



- In order to flow the rated flux in the transformer core applying rated voltage and rated frequency with the help of variac on LV side.
- The HV side of the transformer is kept open.
- After the reading the rated LV side voltage , note down the all meter readings.

2.9.2 SHORT CIRCUIT TEST ON SINGLE PHASE TRANSFORMER

- This test is performed by short-circuiting one winding (usually the low-voltage winding) and applying a reduced voltage to the other winding, as shown in Figure.
- Short circuit test is generally carried out by energizing the HV side with LV side shorted.
- Voltage applied is such that the rated current flows in the windings.
- Voltmeter, ammeter and the wattmeter readings are noted corresponding to the rated current of the windings.



2.10 SYNCHRONOUS GENERATOR

The synchronous generator works on the principle of Faraday laws of electromagnetic induction. Electromagnetic induction states that electromotive force (e.m.f) induced in the armature coil if it is rotating in the uniform magnetic field.

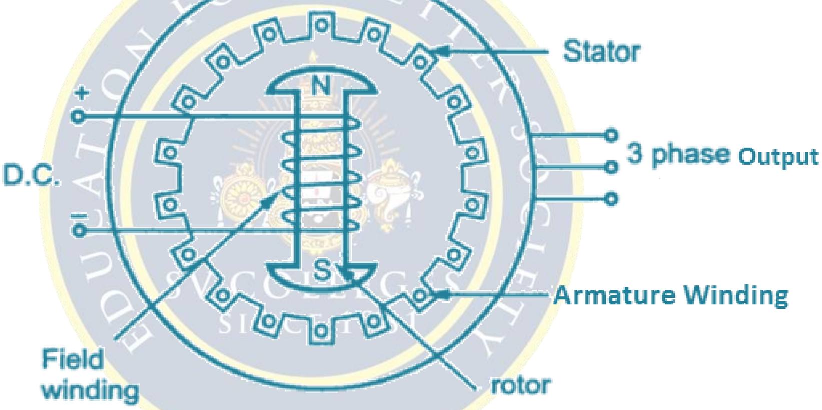
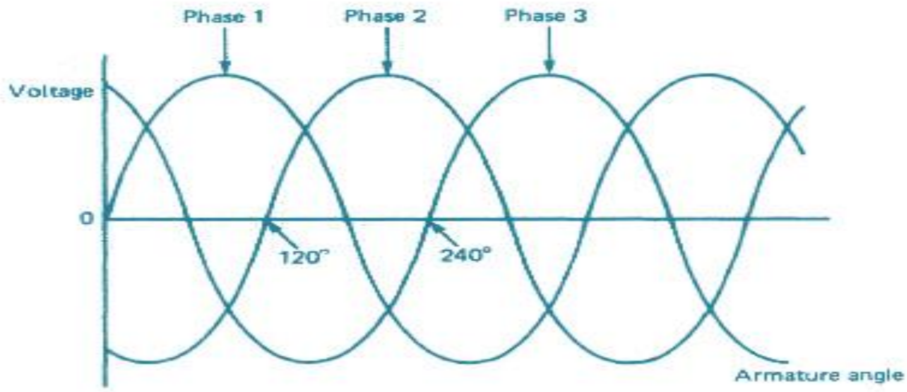


FIG: SYNCHRONOUS GENERATOR OR ALTERNATOR

Generally in practical construction of synchronous generator, armature conductors are stationary and field magnets rotate between them. The rotor of an alternator or a synchronous generator is mechanically coupled to the shaft or the turbine blades, which on being made to rotate at synchronous speed Ns under some mechanical force results in magnetic flux cutting of the stationary armature conductors housed on the stator.

As a direct consequence of this flux cutting an induced emf and current starts to flow through the armature conductors which first flow in one direction for the first half cycle and then in the other direction for the second half cycle for each winding with a definite time lag of 120° due to the space displaced arrangement of 120° between them as shown in the below figure.

This particular phenomena result in 3φ power flow out of the alternator which is then transmitted to the distribution stations for domestic and industrial uses



9. Practice Quiz

1. Which of the following is a function of the brush [c]
a) To convert ac to dc b) To convert dc to ac
c) To collect current and deliver to the load d) None

2. Which of the following is a function of the commutator in a DC generator? [a]
a) To act as a rectifier b) To act as an inverter
c) To act as a junction box for connecting the armature winding ends
d) None of these

3. The direction of generated emf is determined by [d]
a) Lenz's law b) Faraday's laws of electromagnetic induction
c) Fleming's left-hand rule d) Fleming's right-hand rule

4. Which of the following gives the expression for the generated voltage in a dc generator? [c]
a) 4.44fNɸm b) 4.44fTΦm c) ΦNPZ / 60A d) None of these

5. What is the purpose of testing a transformer? [d]
a) Determine its parameters b) Determine its efficiency
c) Determine its regulation d) All of these

6. The short-circuit test is conducted to determine [a]
a) Power factor at a short circuit b) Series parameters
c) Efficiency d) Regulation

7. Maximum efficiency occurs in a transformer when [c]
a) Constant loss > Variable loss b) Constant loss < Variable loss
c) Constant loss = Variable loss d) None of these

8. During open circuit test of a transformer [a]
a) Primary is supplied rated voltage
b) Primary is supplied full load current
c. Primary is supplied current at reduced voltage
d. Primary is supplied rated kVA

9. Which of the following is not standard voltage for power supply in India [d]

a) 11Kv b) 33kV c) 66 kV d) 122 kV

10. Synchronous Speed (Ns) [a]

a) 120f/P b) 120P/f c) 120fp d) None

10. Assignments

S.No	Question	BL	CO
1	Explain about constructional details and working of DC Generator with neat sketch	2	1
2	Explain the procedure to calculate efficiency and regulation of Single-Phase Transformer by conducting OC & SC Tests.	2	1
3	Derive the Expression for EMF equation of DC Generator.	2	1
4	Explain the speed control methods of DC Motor.	2	1
5	Explain the principle and operation of Three Induction Motor.	3	1
6	Explain the principle and operation of Synchronous generator.	3	1

11. Supportive Online Certification Courses

1. A NPTEL Course on Electrical Machines By Prof. G Bhuvaneshwari, Department of Electrical Engineering IIT Delhi
2. A NPTEL Course on Electrical Machines – II by Prof. Tapas Kumar Bhattacharya, Department of Electrical Engineering IIT Delhi.

12. Real Time Applications

S.No	Application	CO
1	The salient pole type motor is generally used for low-speed operations of around 100 to 400 rpm, and they are used in power stations with hydraulic turbines or diesel engine	1
2	Cylindrical rotors are used in high speed electrical machines, usually 1500 RPM to 3000 RPM employed in steam turbine driven alternators like turbo generators	2
3	Synchronous generators are commonly used for variable speed wind-turbine applications, due to their low rotational synchronous speeds that produce the voltage at grid frequency. ... Using an AVR for the excitation of the field voltage, the output voltage of the synchronous generator can be controlled.	3
4	The single phase motor finds its application in vacuum cleaners, fans, washing machines, centrifugal pumps, blowers, washing machines, etc. requirements in one of the most regulated business sectors.	5

13. Contents Beyond the Syllabus

1. Graphical evaluation of frequency response
2. Economics of power supply system

14. Prescribed Text Books & Reference Books

Text Books:

1. D. P. Kothari and I. J. Nagrath - "Basic Electrical Engineering" - Tata McGraw Hill -2010.
2. V.K. Mehta & Rohit Mehta, "Principles of Power System" – S.Chand – 2018.

References:

1. L. S. Bobrow - "Fundamentals of Electrical Engineering" - Oxford University Press – 2011.
2. E. Hughes - "Electrical and Electronics Technology" - Pearson - 2010.
3. C.L. Wadhwa – "Generation Distribution and Utilization of Electrical Energy", 3rd Edition, New Age International Publications.

15. Mini Project Suggestion

1. **Temperature Controlled System:**
   This project shows a temperature-controlled system. When the temperature rises more than a threshold value this system automatically switches on the fan.
2. **Conversion of Single Phase to Three Phase Supply:**
   This paper shows a converter that converts the single-phase supply into a three-phase supply using thyristors. Go through the paper for more information.
3. **Smoke Detector Alarm Circuit:**
   Here is the circuit showing a smoke detector alarm. This can also be used to indicate the fire. This circuit uses a smoke detector and an LM358 Comparator.
4. **Railway Security System based on Wireless Sensor Networks:**
   This paper describes different methods for detecting the defects in railway tracks and methods for maintaining the track are also proposed.
5. **Control Electrical Devices From your android phone:**
   This paper shows the controlling of electrical devices from an android phone using an app. This project uses Arduino for controlling the devices.